Cambridge (CIE) A Level Business

Your notes

Inventory Management

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- Methods of Inventory Management
- Just in Time

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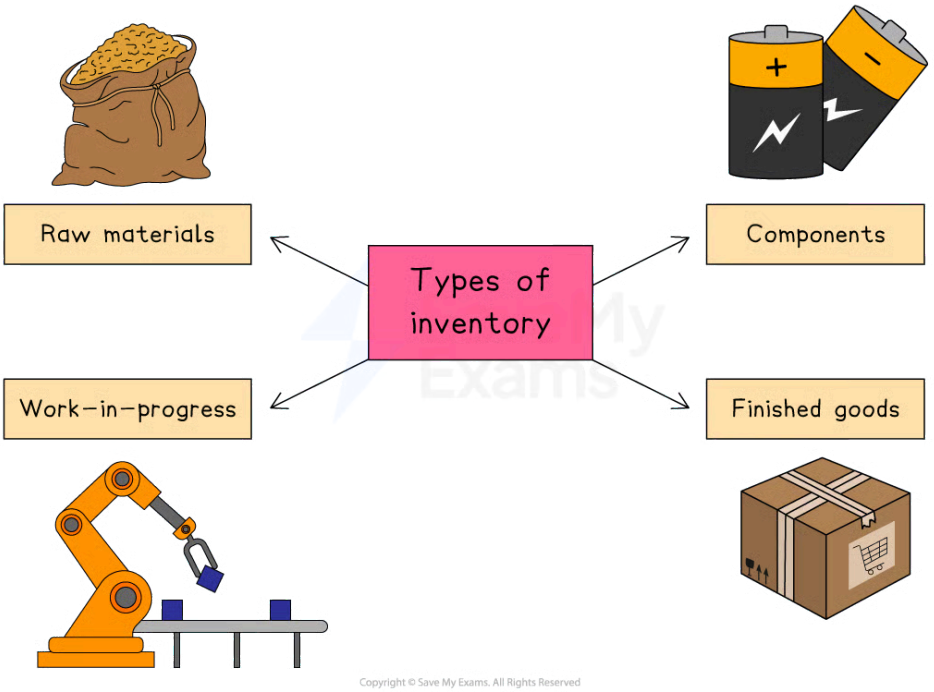
The purpose of inventory

- **Inventory** is the goods and materials a business holds for the ultimate goal of resale or for use in production

Types of inventory



Inventory consists of raw materials, components, work-in-progress and finished goods

Raw materials	Components
▪ These are the natural resources or goods that are **used to make other products** ▪ Examples in vehicle production include steel, plastic and oil	▪ The parts used in the production of goods ▪ Examples in vehicle production include circuit boards, body panels and gearboxes
Work-in-progress	**Finished goods**

▪ This includes **partially finished** products awaiting completion
  ▪ In car production, this could include vehicles awaiting spray-painting or quality control checks
	▪ These are completed products which require no further manufacturing
	  ▪ In car production, these are vehicles ready for sale to the final customers

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Costs and benefits of holding inventory

- The management of stock is an important consideration for businesses
- Problems may arise from holding inventory
  - Storage costs (e.g. warehouse rental, security costs) will be higher than necessary
  - The risk of spoilage and stock shrinkage is increased, leading to increased costs
  - Money tied up in stock could be used elsewhere in the business
  - Time and resources are needed to monitor and track stock levels
- However, if a business holds inventory, it means that
  - Products are available when needed, improving customer satisfaction
  - Buying quantities of inventory and storing it may reduce unit costs due to bulk buying discounts
  - Production isn't held up by supply issues
  - Extra stock helps meet peaks in sales, such as during holiday periods

The importance of supply chain management

- The **supply chain** is the network of organisations, people, activities and resources that move a product **from its basic raw material right through to the final customer**
- It includes the following:
  - **Stock control**: planning, implementing and monitoring the movement of raw materials, components, work-in-progress and finished goods
  - **Quality control**: ensuring output meets standards so that the end product is safe and meets customer expectations
  - **Transport networks**: ensuring efficient deliveries of goods to customers, taking account of speed, reliability and costs
  - **Supplier networks**: developing strong relationships with suppliers willing to work collaboratively to improve quality
- **Global supply chains** require these activities to be coordinated **across international borders**
  - Some stages can be **completed at lower cost** in certain countries

- **E.g. China** has a reputation for producing high quality, low-cost electronics components
- Labour-intensive processing, such as clothes manufacturing, is outsourced to countries with **low labour costs**, such as **Vietnam**
- **Scarce raw materials** may only be available in certain countries/regions

Your notes

Building an effective supply chain

Decision area	What it involves
Make or buy (produce in-house or outsource)	▪ Decide whether to produce in-house or buy from specialists
Choosing suppliers	▪ Single, trusted supplier or multiple backup options ▪ Local or global sources
Purchasing approach	▪ Bulk buying for discounts or frequent, small orders for flexibility
Information sharing and use of technology	▪ Real-time data, bar codes, RFID and cloud systems keep production partners fully informed
Logistics structure	▪ How many and where warehouses are needed ▪ Transport options — use an external haulier or keep control of deliveries

Why does an effective supply chain matter?

1. **Faster delivery to customers**
   - When every link is well-coordinated, products move quickly from factory to shelf, beating slower rivals
2. **Lower operating costs**
   - Just-in-time (JIT) deliveries and bulk purchasing agreements reduce storage, handling and material expenses
   - E.g. **Aldi** negotiates long-term contracts with a small group of trusted suppliers, helping it minimise warehouse inventory and keep prices low
3. **Consistent quality**
   - Close, long-term relationships with suppliers make it easier to enforce standards and fix problems early
4. **Greater resilience to shocks**
   - A well-planned supply chain includes backup suppliers and effective data sharing so the business can adapt when something goes wrong

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Inventory control charts

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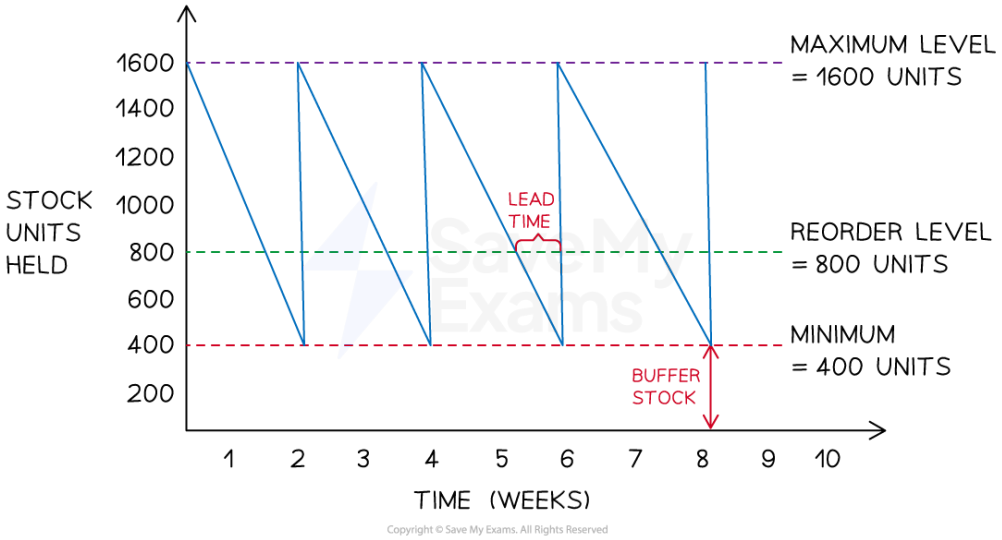
- An **inventory control diagram** shows how inventory moves into and out of a business over time

An inventory control diagram

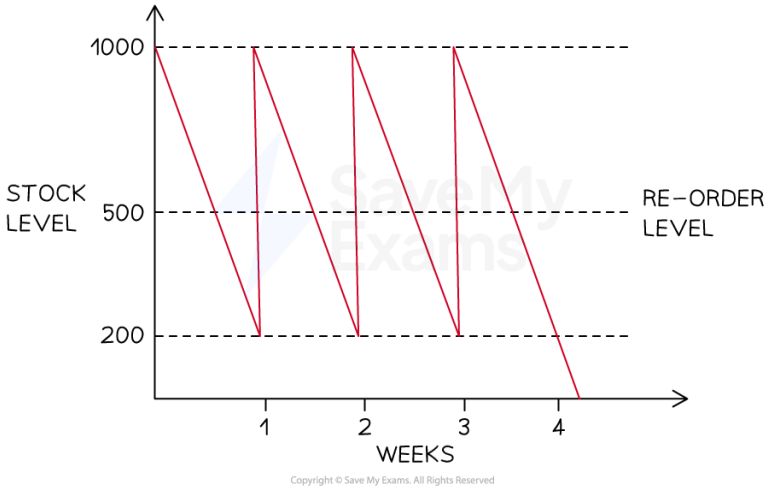


An example of an inventory control diagram for a small manufacturing business

- The **maximum inventory level** is the maximum amount of stock a business is able to hold in normal circumstances (1,600)
- The **reorder level** is the level at which a business places a new order with its supplier (800)
- The **minimum stock level** is also known as the **buffer inventory** level and is the lowest level to which a business is willing to allow inventory levels to fall (400)
- **Lead time** is the length of time from the point of inventory being ordered from the supplier to it being delivered (one week)
- The stock level line shows how inventory levels change over the given time period
  - As inventory is used up, a downwards slope is plotted
  - When an order is delivered by a supplier, the stock level line shoots upwards

Worked Example

The diagram below shows inventory movements of kitchen shelving units sold by TamFix Limited.



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Identify the following points:

1. The minimum stock level
2. The reorder level
3. The reorder quantity
4. The lead time for kitchen shelving units

(4)

Answer:

Step 1: Identify the minimum stock level

- The minimum stock level is identified by the bottommost dotted line
- In this case, it shows that the minimum stock level is 200 units **(1)**

Step 2: Identify the reorder level

- The reorder level is clearly identified on the diagram
- In this case, it shows that the reorder level is 500 units **(1)**

Step 3: Identify the reorder quantity

- The reorder quantity is the difference between the maximum stock level (shown by the topmost dotted line) and the minimum stock level

$$= 1000 \text{ units} - 200 \text{ units}$$

$$= 800 \text{ units}$$

- The reorder quantity is, therefore, 800 units **(1)**

Step 4: Identify the lead time for kitchen shelving units

- The lead time is the difference in time between the time an order is placed and the time the inventory is delivered
- In this case, assuming a five-day working week, the lead time for shelving units is two days **(1)**

Examiner Tips and Tricks

Always clearly label lead time, reorder level and buffer stock, and include units

Show your workings — examiners award method marks even if the final figure is wrong

Your notes

Buffer inventory

- Buffer inventory is a quantity of stock kept in case of shortages
  - This can provide a **competitive edge** over rivals unable to meet demand
  - This approach is commonly called **'just in case'** stock control
- The decision to keep buffer stocks is one that businesses have to weigh up very carefully
  - The decision will be influenced by the nature of the business and the product/service it provides

Evaluating the use of buffer stocks

Advantages	Disadvantages
▪ **Stability in supply** Buffer stocks ensure a stable supply of goods which is able to respond to unexpected customer demand ▪ **Price stabilisation** Buffer stocks can help prevent extreme price fluctuations, as it helps the market **avoid shortages,** which would result in rapid price increases ▪ **Raw materials security** Businesses that are **dependent on particular raw materials** avoid disruption to their supply ▪ **Competitive advantage** By having a reliable supply of goods, businesses can **gain a reputation** for always being able to meet the needs of their customers	▪ **Cost** Holding buffer stocks can be expensive, as it requires storage facilities and **inventory management systems** ▪ **Risk of obsolescence** Buffer stocks can become obsolete if the demand for a particular product or input declines ▪ **Opportunity cost** Holding buffer stocks **ties up capital** that could be invested in other areas of the business

Your notes

Just in Time versus Just in Case

1. The just-in-time approach (JIT)

- The **just-in-time (JIT)** approach involves **raw materials** and **components** being **ordered as required** and delivered "just in time" to be used in production
- Raw materials and components are ordered from a **small number** of **trusted key suppliers just before they are to be used**
  - **Close**, **long-term** relationships with these suppliers need to be developed
    - Many businesses using JIT inventory management systems aim to source raw materials and components from local or regional suppliers
    - They must be **flexible** and **reliable**
    - They may be required to **hold inventory on behalf** of a JIT-operating customer
    - They are often in **close proximity** to their key JIT-operating customer

Evaluating the just-in-time approach

Advantages	Disadvantages
▪ **Stockholding costs**, including storage rental and security, are minimised ▪ Inventory and finished goods are **less likely to be damaged** in storage ▪ Close working relationships are developed with a **small number of trusted suppliers** ▪ **Cash flow is improved**, as money that is not tied up in stocks **can be put to other uses** ▪ **Unused storage space** is available for productive use or can be disposed of	▪ Bulk buying **economies of scale** are not generally possible ▪ Unable to respond to **unexpected increases in demand** without precise forecasting of demand ▪ High **administration costs** due to frequent ordering of stock ▪ **Unreliable suppliers** (e.g. late or poor-quality deliveries) can quickly halt production ▪ **External factors** can delay delivery of stock, e.g. increased border checks on imported goods since Brexit

Examiner Tips and Tricks

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When you explain the JIT approach, link every benefit (less inventory, lower storage costs) to a risk (stockouts if deliveries slip)

Showing both sides proves you understand why lean systems boost profit only when suppliers are rock-solid

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2. The just-in-case approach

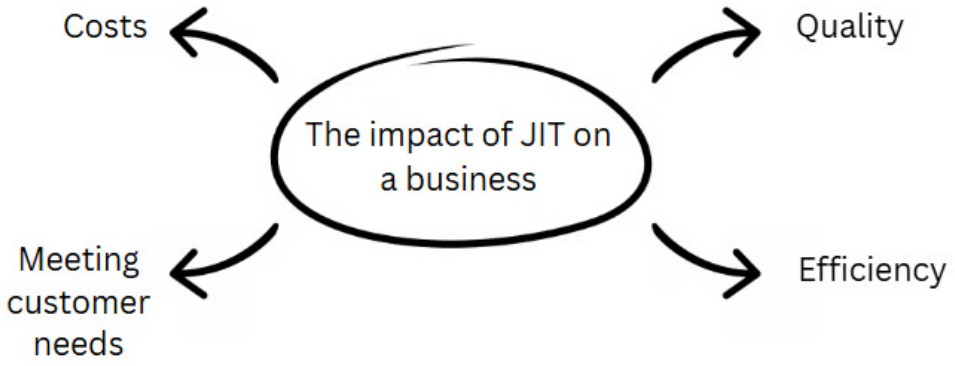
- **Just-in-case inventory management** involves a business holding a quantity of raw materials, components or finished goods as **buffer stock**
  - Stock is held in case of **shortages** so as to provide a **competitive edge** over rivals unable to meet demand
- The decision to keep buffer stocks is one that businesses have to weigh up very carefully
  - Holding inventory incurs **storage and security costs** and can increase **waste**, as stock could be damaged, stolen or become **obsolete**
  - Failing to hold enough inventory could mean a business is **unable to meet demand**, potentially missing out on **sales revenue**

Evaluating the just-in-case approach

Advantages	Disadvantages
▪ Buffer stocks ensure a **stable supply of goods,** allowing a business to respond to increases in demand ▪ Extreme **price fluctuations** due to shortages of inventory can be avoided ▪ Businesses that are **dependent on particular raw materials** avoid supply disruption ▪ Businesses with a regular supply **gain a reputation** for always being able to meet customers' needs	▪ Holding buffer stocks can be expensive, as it requires storage facilities and **inventory management systems** ▪ Buffer stocks can become **obsolete** if the demand for a particular product or input declines ▪ Holding buffer stocks **ties up cash** that could be invested in other areas of the business

The impact of adopting a JIT approach

- The JIT approach has significant effects on several areas of a business



Adopting JIT can have a significant impact on costs, quality, efficiency and a business's ability to meet customer needs

Your notes

Impact on costs

- **Lower inventory costs**
  - Since stock is only ordered when needed, businesses spend less on storage facilities and insurance
- **Less money tied up in stock**
  - Firms have more working capital available for other uses, like marketing or staff training
- **Fewer waste costs**
  - JIT reduces the risk of obsolete or damaged stock, especially for perishable or seasonal items

Impact on quality

- **Improved supplier quality**
  - Businesses work closely with reliable suppliers to ensure materials meet high quality standards
- **Faster feedback loops**
  - Any quality issues are noticed and fixed quickly, as production is continuous and lean
- **Focus on right-first-time**
  - As there's no buffer stock, production teams must work efficiently and avoid errors

Impact on efficiency

- **Streamlined production**
  - JIT promotes smoother workflows and reduces idle time between stages of production

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- **Motivated workforce**
  - Workers often have more responsibility and take part in continuous improvement (Kaizen)
- **Space-saving**
  - With fewer storage needs, factory space is better used for production activities

Impact on meeting customer needs

- **Faster response to changes**
  - Businesses can adapt quickly to shifts in consumer demand, reducing the risk of overproducing unpopular products
- **Customisation**
  - JIT allows more flexible and tailored production, often linked to modern mass customisation systems.
- **Freshness and quality**
  - Products are made closer to the time of sale, especially useful in food and fashion industries

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